

TECHBIG SOLUTIONS

Ambattur, Chennai

Microsoft SQL Server – Complete E2E Training Syllabus

T-SQL | Database Development | DBA | Performance | HA/DR | BI Integration

Program Objective: Build complete practical SQL Server capability covering T-SQL, database design, stored procedures, functions, triggers, transactions, indexing, performance tuning, security, backup/recovery, high availability, automation, administration and real-world projects.

Reference basis: Microsoft SQL Server documentation and learning resources covering Database Engine, T-SQL, security, backup/recovery, performance, SQL Server Agent, high availability and Azure SQL.

Program	Microsoft SQL Server – Complete E2E
Level	Beginner → Advanced SQL Developer / DBA / BI Professional
Core Areas	T-SQL Database Design Stored Procedures Security Administration
Advanced Areas	Performance Backup/Restore HA/DR Automation Azure SQL BI
Suggested Duration	120–180 hours, customizable by batch and project depth

1. SQL Server Overview & Database Concepts

- SQL Server architecture and editions overview
- Relational database concepts
- Database engine, instances and databases
- Tables, schemas, views and objects
- OLTP vs OLAP
- Development, test and production environments

2. SQL Server Installation & Configuration

- SQL Server installation concepts
- Instance configuration
- Database Engine services
- Authentication modes
- Server and database configuration
- SQL Server Configuration Manager
- SSMS installation and connectivity

3. SQL Server Management Studio (SSMS)

- SSMS interface and Object Explorer
- Query Editor
- Registered servers and connections
- Execution plans and query tools
- Import/export and scripting
- SQL Server Agent interface
- Database object scripting and documentation

4. T-SQL Fundamentals

- SELECT statements
- WHERE, ORDER BY and DISTINCT
- Aliases and expressions
- Comparison and logical operators
- TOP and pagination
- Comments and coding standards

5. SQL Server Data Types

- Integer and decimal types
- Character and Unicode types
- Date/time types
- Binary and uniqueidentifier
- XML and JSON data types
- NULL handling
- Choosing appropriate data types

6. Database & Table Design

- CREATE/ALTER/DROP database
- Schemas
- CREATE/ALTER/DROP TABLE
- Primary keys and foreign keys
- Constraints
- Identity columns
- Default and check constraints
- Normalization and data integrity

7. Data Manipulation Language (DML)

- INSERT
- UPDATE
- DELETE
- MERGE concepts
- OUTPUT clause
- Transactions with DML
- Safe update/delete practices

8. Querying Multiple Tables

- INNER, LEFT, RIGHT and FULL joins
- CROSS JOIN
- Self joins
- Join conditions and duplicates
- Subqueries
- Correlated subqueries
- EXISTS and NOT EXISTS
- APPLY operator

9. Aggregation & Grouping

- COUNT, SUM, AVG, MIN and MAX
- GROUP BY
- HAVING
- Conditional aggregation
- GROUPING SETS
- ROLLUP and CUBE
- Aggregate query optimization

10. Advanced SELECT & Query Techniques

- CASE expressions
- Common table expressions (CTEs)
- Recursive CTEs
- UNION, UNION ALL, INTERSECT and EXCEPT
- PIVOT and UNPIVOT
- Windowed result-set patterns
- Advanced filtering

11. Views & Synonyms

- Creating and modifying views
- Updatable views
- Indexed-view concepts
- Security through views
- Synonyms
- View performance and dependencies

12. Stored Procedures

- Creating stored procedures
- Input and output parameters
- Return values
- Dynamic SQL
- Temporary tables and table variables
- Error handling in procedures
- Procedure security and deployment

13. User-Defined Functions

- Scalar functions
- Inline table-valued functions
- Multi-statement TVFs
- Parameters and return types
- Function limitations
- Performance considerations

14. Triggers

- DML triggers

- AFTER and INSTEAD OF triggers
- Inserted/deleted logical tables
- DDL triggers overview
- Trigger design and recursion
- Auditing and business-rule use cases
- Trigger performance considerations

15. Temporary Objects & Table Variables

- Local and global temporary tables
- Table variables
- Temporary object indexing
- Scope and lifecycle
- TempDB considerations
- Choosing temporary tables vs table variables

16. Transactions & Concurrency

- ACID principles
- BEGIN/COMMIT/ROLLBACK
- TRY/CATCH transactions
- Isolation levels
- Locks and blocking
- Deadlocks
- Snapshot isolation
- Transaction troubleshooting

17. Indexing

- Clustered and nonclustered indexes
- Included columns
- Composite indexes
- Filtered indexes
- Unique indexes
- Columnstore indexes overview
- Index fragmentation
- Index maintenance and design

18. Query Optimization & Execution Plans

- Estimated vs actual execution plans
- Scan vs seek
- Join algorithms
- Statistics
- Cardinality estimation
- Parameter sniffing
- Query Store overview
- Performance tuning methodology

19. SQL Server Security

- Logins and users
- Server roles and database roles
- GRANT, DENY and REVOKE
- Object-level permissions
- Schemas and ownership
- Application security
- Row-level security overview
- Auditing and security best practices

20. Authentication & Encryption

- Windows and SQL authentication
- Contained database users
- Transparent Data Encryption overview
- Always Encrypted overview
- TLS and encrypted connections
- Credential and secret management
- Security assessment concepts

21. Backup & Restore

- Full, differential and transaction-log backups
- Recovery models
- Point-in-time recovery
- Backup strategies
- Restore sequences
- Backup verification
- Disaster-recovery planning

22. High Availability & Disaster Recovery

- Always On availability groups overview
- Failover concepts
- Log shipping
- Database mirroring legacy concepts
- Failover clustering overview
- RPO and RTO
- HA/DR architecture and testing

23. SQL Server Agent & Job Automation

- Jobs, steps and schedules
- Alerts and operators
- Job history
- Maintenance jobs
- Database backup automation
- Failure notifications

- Production scheduling and monitoring

24. SQL Server Maintenance

- DBCC commands overview
- Statistics maintenance
- Index maintenance
- Integrity checks
- TempDB configuration concepts
- Database consistency
- Maintenance plans and custom maintenance scripts

25. Import, Export & Data Integration

- Import/Export Wizard
- BCP and BULK INSERT
- OPENROWSET concepts
- CSV/Excel ingestion
- SSIS overview
- ETL concepts
- Data validation and reconciliation

26. JSON, XML & Modern SQL Server Data

- FOR JSON
- OPENJSON
- JSON_VALUE and JSON_QUERY
- XML data type
- XML methods
- Semi-structured data processing
- API/database integration scenarios

27. Reporting & BI Integration

- Reporting Services (SSRS) overview
- Power BI integration concepts
- Views/stored procedures for reporting
- Operational vs analytical reporting
- Data marts and dimensional modeling
- Secure reporting data access

28. Monitoring & Troubleshooting

- SQL Server logs
- Activity Monitor
- DMVs and DMFs
- Blocking and deadlock analysis
- CPU, memory and I/O analysis
- Wait statistics
- Query Store monitoring

- Incident and root-cause analysis

29. SQL Server Administration & Governance

- Database lifecycle management
- Capacity planning
- Storage management
- Configuration baselines
- Change management
- Patch/upgrade concepts
- Documentation and operational standards

30. Cloud & Azure SQL Overview

- Azure SQL Database overview
- Azure SQL Managed Instance overview
- SQL Server on Azure VM
- Migration approaches
- Connectivity and security
- Backup/DR in Azure
- Hybrid SQL Server environments

31. Development Best Practices

- T-SQL coding standards
- Naming conventions
- Reusable SQL components
- Parameterized queries
- Error handling and logging
- Source control concepts
- Deployment scripts and rollback
- Code review practices

32. Real-Time E2E Projects

- Employee/HR database design with security and reporting
- Sales and customer management database with stored procedures
- Inventory database with transactions, indexes and audit triggers
- Finance database with reconciliation and reporting queries
- ETL pipeline: CSV/Excel → SQL Server → reporting layer
- Performance-tuning project using execution plans and Query Store
- Backup/restore and point-in-time recovery project
- High-availability design using Always On concepts
- Power BI reporting solution over SQL Server
- Complete production SQL Server administration and support project

33. Production Support & Troubleshooting

- Database connectivity failures
- Login and permission issues

- Blocking and deadlocks
- Slow-query incidents
- Space/storage issues
- Backup/restore failures
- Job failures
- Data corruption/consistency investigation
- Performance degradation
- Root-cause analysis and incident management

34. Interview & Certification Preparation

- SQL and T-SQL interview questions
- Complex query exercises
- Joins, CTEs and window-function scenarios
- Stored procedure and trigger questions
- Indexing and performance case studies
- Backup/restore and HA/DR questions
- Security and administration scenarios
- Real-time project explanation
- Mock practical assessment and certification preparation

Recommended Hands-on Deliverables

- Design and implement a normalized SQL Server database with keys, constraints, relationships and security.
- Develop complex T-SQL using joins, CTEs, window functions, aggregation and advanced query techniques.
- Build stored procedures, functions, triggers and transaction-safe business processes.
- Diagnose a slow production query using execution plans, indexes, statistics and Query Store.
- Implement backup, restore and point-in-time recovery procedures and document an HA/DR strategy.
- Build an end-to-end SQL Server reporting/BI solution and establish production monitoring and support procedures.

Training Outcome

Learners completing this program should be able to develop and administer SQL Server databases, write production-quality T-SQL, optimize workloads, implement security and recovery strategies, automate operational tasks, integrate reporting/BI solutions and troubleshoot production database incidents.

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